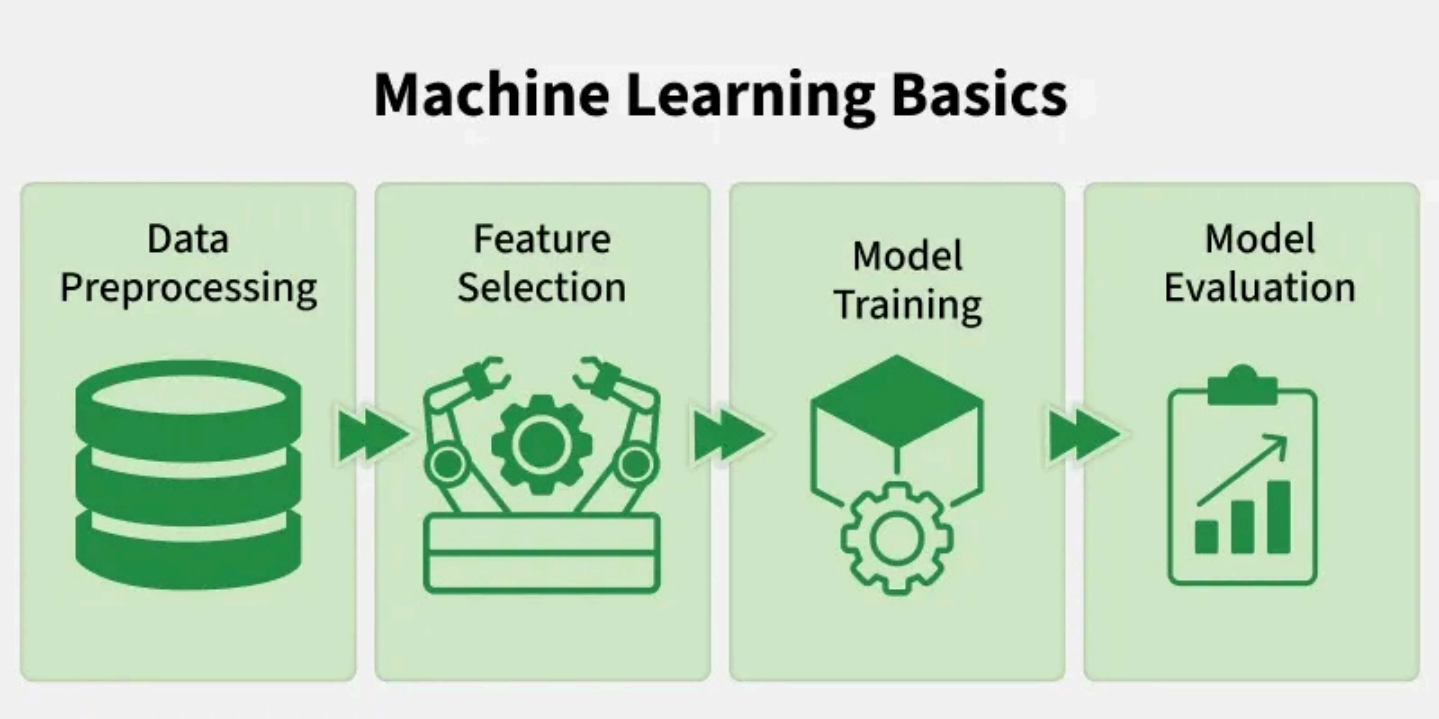
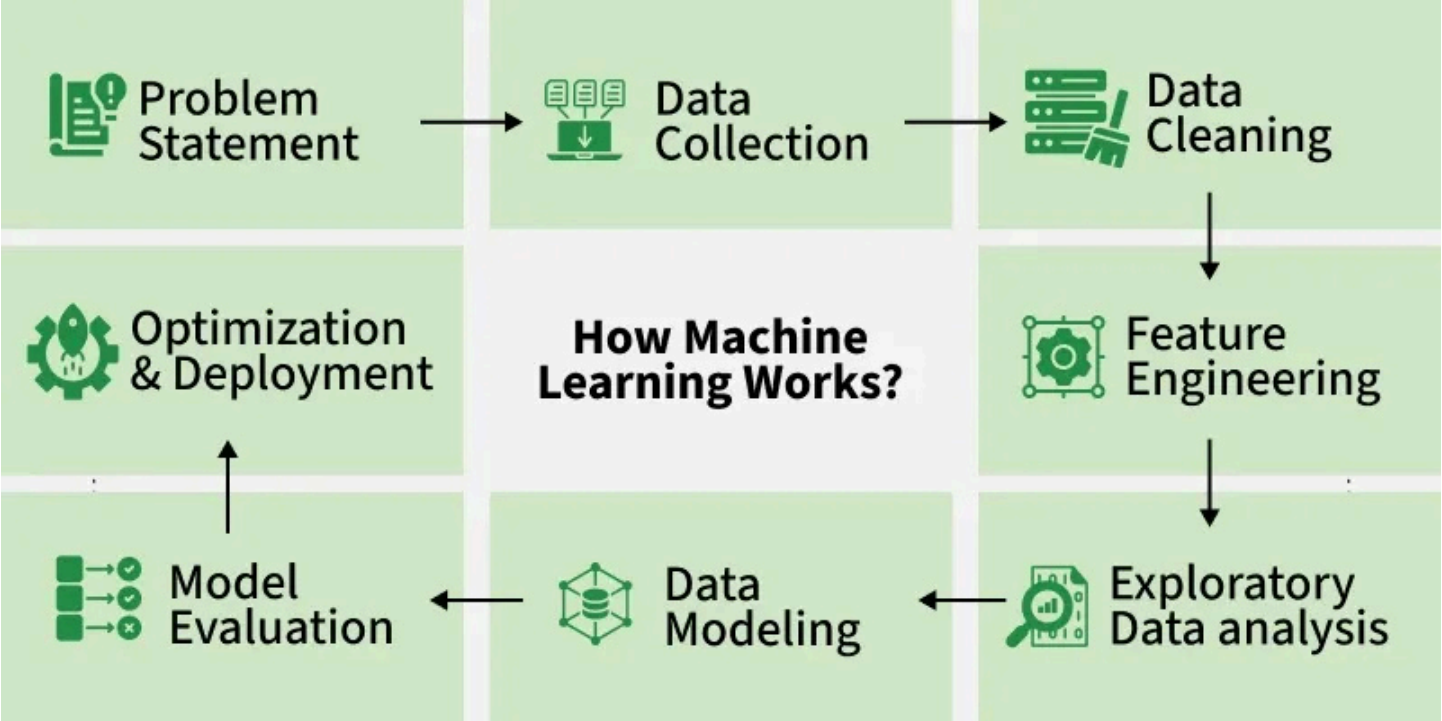
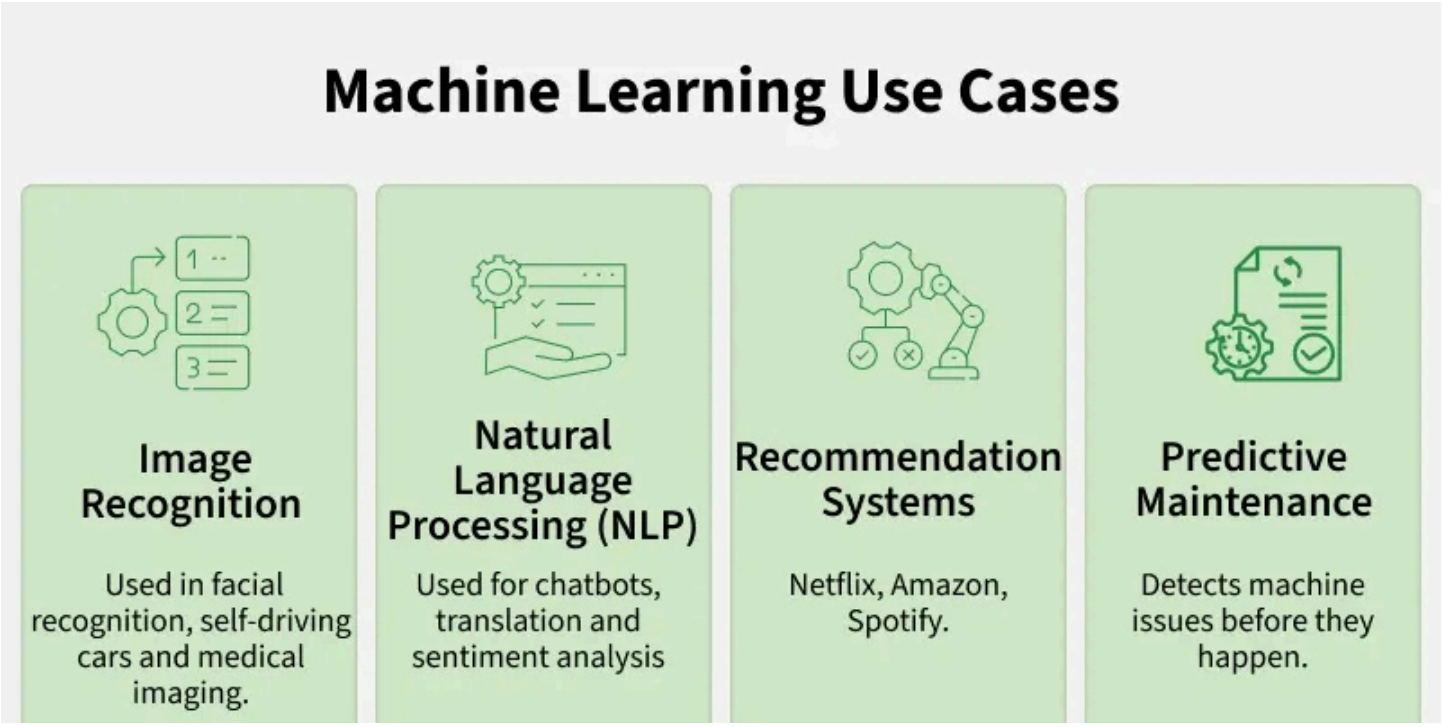
MACHINE LEARNING

What is Machine Learning

Machine Learning is a subset of Artificial Intelligence that enables systems to automatically learn and improve from experience without being explicitly programmed. It focuses on developing algorithms that can analyze data, identify patterns, and make decisions.







Types

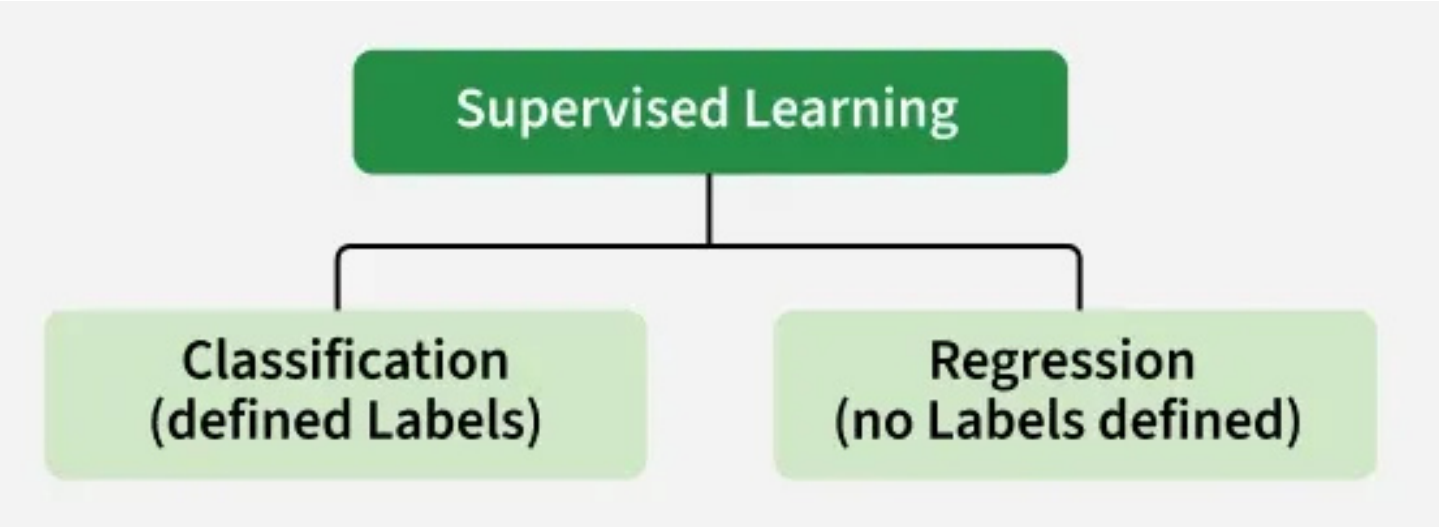
Machine Learning is mainly divided into three core types:

- **Supervised Learning:**Supervised learning is a type of machine learning where a model learns from labelled data, meaning each input has a correct output. The model compares its predictions with actual results and improves over time to increase accuracy.

Its main features are:

- Each input has a known output
- Adjusts itself to reduce prediction errors
- Make accurate predictions on new data
- For example it recognizing handwritten digits from trained data
- Each input has a known

Types of Supervised Learning



Classification: Where the output is a categorical variable (e.g., spam vs. non-spam emails, yes vs. no).

Regression: Where the output is a continuous variable (e.g., predicting house prices, stock prices).

- **Unsupervised Learning :** Unsupervised Learning is a type of machine learning where the model works without labelled data. It learns patterns on its own by grouping similar data points or finding hidden structures without any human intervention.
- It is used for tasks like clustering, dimensionality reduction and Association Rule Learning.
- Helps identify hidden patterns in data
- Useful for grouping, compression and anomaly detection

- ▢ **Reinforcement Learning**: Learns through trial and error to maximize rewards, ideal for decision-making tasks.Reinforcement Learning revolves around the idea that an agent (the learner or decision-maker) interacts with an environment to achieve a goal. The agent performs actions and receives feedback to optimize its decision-making over time.
- Agent: The decision-maker that performs actions.
- Environment: The world or system in which the agent operates.
- State: The situation or condition the agent is currently in.
- Action: The possible moves or decisions the agent can make.
- Reward: The feedback or result from the environment based on the agent's action.

Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for classification problems. Unlike linear regression, which predicts continuous values it predicts the probability that an input belongs to a specific class.

- It is used for binary classification where the output can be one of two possible categories such as Yes/No, True/False or 0/1.
- It uses sigmoid function to convert inputs into a probability value between 0 and 1.

Types of Logistic Regression

Logistic regression can be classified into three main types based on the nature of the dependent variable:

1. **Binomial Logistic Regression**: This type is used when the dependent variable has only two possible categories. Examples include Yes/No, Pass/Fail or 0/1. It is the most common form of logistic regression and is used for binary classification problems.
2. **Multinomial Logistic Regression**: This is used when the dependent variable has three or more possible categories that are not ordered. For example, classifying animals into categories like "cat," "dog" or "sheep." It extends the binary logistic regression to handle multiple classes.
3. **Ordinal Logistic Regression**: This type applies when the dependent variable has three or more categories with a natural order or ranking. Examples include ratings

like "low," "medium" and "high." It takes the order of the categories into account when modeling

Linear Regression

Linear Regression is a fundamental supervised learning algorithm used to model the relationship between a dependent variable and one or more independent variables. It predicts continuous values by fitting a straight line that best represents the data.

- It assumes that there is a linear relationship between the input and output
- Uses a best-fit line to make predictions
- Commonly used in forecasting, trend analysis, and predictive modelling

https://github.com/gourisatheesh00-web/AI-internship